

Scientific Programming with Python

Assignment: Matplotlib/Seaborn and Data Visualization

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Goal The purpose of this assignment is to become familiar with the matplotlib and/or seaborn modules, explore some data that you are interested in, and to put into practice the scientific programming concepts learned in lecture so far. This assignment will push you to work efficiently, manage your time well, develop a notebook in a concise manner, and to communicate clearly.

Problem and Input Data For this assignment you are free to choose a topic and dataset that you are interested to explore, and then convey what you find out to your peers. This data could be something that you are already working with, or something entirely new. If you want to explore something new, the follow website are possible resources to explore:

- Kaggle <https://www.kaggle.com> [1]
- Data.gov <https://data.gov> [2]
- Google Dataset Search <https://datasetsearch.research.google.com> [3]
- 538 Datasets <https://github.com/fivethirtyeight/data> [4]
- Awesome Public Datasets <https://github.com/awesomedata/awesome-public-datasets.git> [5]

IMPORTANT: When choosing a dataset, keep it respectful. Do not choose something or discuss the results such that is going to be offensive to others (e.g., racial, gendered, violent). As adults, I will let you decide what might or might not be offensive to your peers.

Assignment Task

Task 1 Using the Python tools covered so far in class (e.g., Pandas, NumPy, Matplotlib), explore your chosen dataset. Your homework solution must demonstrate a reasonable effort to explore your data.

Dataset Requirements:

The dataset you investigate must be of a reasonable size since other students will download and run your Jupyter notebook. **You will provide the final dataset as a CSV-formatted file.**

- Small dataset (ca. 0–4 MB): you will submit via LEA.
- Large datasets (ca. 5–50 MB): will require that you provide an accessible web link to it.
- Very large datasets (e.g. hundreds of MB or more): are not allowed.

If you have a very large dataset, you may need to prune (i.e., remove unnecessary columns) or truncate (e.g., via random sampling) the data.

Brainstorming Ideas for Helping

- statistics
- moving averages
- histograms
- coloring data based on its type (e.g., sequential, divergent)
- factorizing data
- correlation analysis
- polynomial fitting

Allowed Python3 [6, 7] functions & libraries/modules

- All built-in functions
- Matplotlib [8], Seaborn [9], NumPy [10], and Pandas [11, 12]
- typing

Assignment Due Turn in your solution as a Jupyter-notebook to **LEA** by **July 6th, 2026 at 13:30**.

Note 1 : Include your input **CSV-formatted** dataset, or a link to it for the peer-review.

Note 2 : Please include your **SciPro ID** at the **top** of your notebook.

Note 3 : You **do not** need to consider **significant figures**.

References

- [1] Sculley, D. and coworkers, Kaggle. Available at <https://www.kaggle.com>. Online; accessed on June 13, 2024.
- [2] United States Government, Data.gov. Available at <https://data.gov>. Online; accessed on June 13, 2024.
- [3] Google LLC, Google Dataset Search. Available at <https://datasetsearch.research.google.com>. Online; accessed on June 13, 2024.
- [4] fivethirtyeight, Data. Available at <https://github.com/fivethirtyeight/data>. Online; accessed on June 13, 2024.
- [5] caesar0301, Awesome Public Datasets. Available at <https://github.com/awesomedata/awesome-public-datasets.git>. Online; accessed on June 13, 2024.
- [6] Python Software Foundation. Python Language Reference, version 3.12. Available at <http://www.python.org>. Online; accessed on May 8, 2024.
- [7] van Rossum, G. Python tutorial, Technical Report CS-R9526, Centrum voor Wiskunde en Informatica (CWI), Amsterdam, 1995.
- [8] Hunter, J.D., 2007. Matplotlib: A 2D graphics environment. IEEE Annals of the History of Computing, 9(03), pp.90-95.
- [9] Waskom, M. L., 2021. seaborn: statistical data visualization. Journal of Open Source Software, 6(60), 3021 (DOI: 10.21105/joss.03021).
- [10] Harris, C.R., Millman, K.J., van der Walt, S.J. et al. Array programming with NumPy. Nature, 585 (2020) 357–362 (DOI: 10.1038/s41586-020-2649-2)
- [11] The Pandas Development Team pandas-dev/pandas: Pandas Zenodo, 2020 (<https://pandas.pydata.org>)
- [12] Pandas user guide, https://pandas.pydata.org/docs/user_guide/index.html#user-guide. Online; accessed on May 8, 2024.